

as simple as lens." Development of powerful LEDs performing to their best in narrow spectres of intense light has exposed the consumer market to another health hazard. A lot of research is going on to study the eye-safety testing and evaluation of LEDs. Companies like Orb Optronix have applied for ANSI- and IESNA-recommended practices to take care of such a situation.

Designing for safety is a concern to both the LED manufacturers as well as product designers. But in optics, LED manufacturers do not have control over how their LED chip will be used. In optics, the total result cannot be determined by combining individual results, which makes testing role of LED-based product manufacturers more critical than LED component manufacturers.

Consumer safety is the key responsibility in product testing. It is

necessary to carefully evaluate every hazardous condition and measure spectral radiance.

Expectations and goals for future

The only way to distinguish between good and bad products is the quality, for which standards and tests are very important. "Testing and qualifications are the market differentiator when so many low-cost, low-quality products enter the market," feels Mahajan. Introduction of cheaper LED lamps by unorganised players has hampered the penetration of good-quality certified LED products. LED lighting is expected to have a great future, but the penetration largely depends on the affordability.

There needs to be some kind of specifications and standardisation as early as possible. BEE, Central Power Research Institute and other organisa-

tions in India are working towards it. At the LED Lighting Congress, Kumar voiced another challenge, "Even though we have standards, we still don't have a national-level testing laboratory. Within this year, the work for the national test facility for LEDs will begin." Kumar also announced an impending global tender for a million LED lamps expected this year for which BEE will announce the specifications.

"When it comes to LED products, the focus is on achieving maximum light output, i.e., maximum number of lumens with minimal electrical consumption. The move is towards getting better luminous efficacy, better thermal management, better life time and colour consistency," says Mahajan. Standardisation and measurement will play an important role in this. ●

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